

Smart Integrated Hospital & Emergency Management System

- Project Proposal

Project Title

- Smart Integrated Hospital & Emergency Management System

Type of Project

- Web Application with AI-Powered Healthcare and Emergency Management Features

Objectives

- Provide a centralized hospital information system
- Manage doctor directories and availability
- Track beds, ICU units, oxygen supply and emergency services
- Enable appointment booking and management
- Store and manage patient medical records
- Provide AI-based healthcare assistance
- Support women, newborn care and vaccination management

Problem Statement

- Difficulty finding accurate hospital resource information
- Medical records scattered across providers
- Finding doctors, ambulances, blood donors and medicines takes time
- Need continuous monitoring for mothers and newborns

Deliverables

- User Management System
- Patient Management Module
- Doctor Management Module
- Hospital Resource Tracking System
- Emergency Services Module
- Appointment Booking System
- Pharmacy Management System
- AI Health Assistant
- Women & Baby Care Management System
- Administrative Dashboard

Methodology

- Agile Software Development
- Requirements Gathering
- System Design
- Frontend & Backend Development
- Testing and Validation
- Deployment and Maintenance

Project Overview

- Centralized healthcare information platform
- Doctor search and appointment booking
- Medical history access
- AI-powered recommendations
- Nearby emergency resource identification

Scope

- Doctor search
- Resource monitoring
- Appointment booking
- Emergency support
- Pharmacy management
- Pregnancy and baby care
- Future GPS tracking and telemedicine support

Technology Stack

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js / PHP
- Database: MySQL
- IDE: Visual Studio Code
- AI Integration: Recommendation Engine

Risks and Dependencies

- Internet connectivity dependency
- Data privacy and security challenges
- Hospital data integration requirements
- Third-party API availability
- Healthcare data accuracy